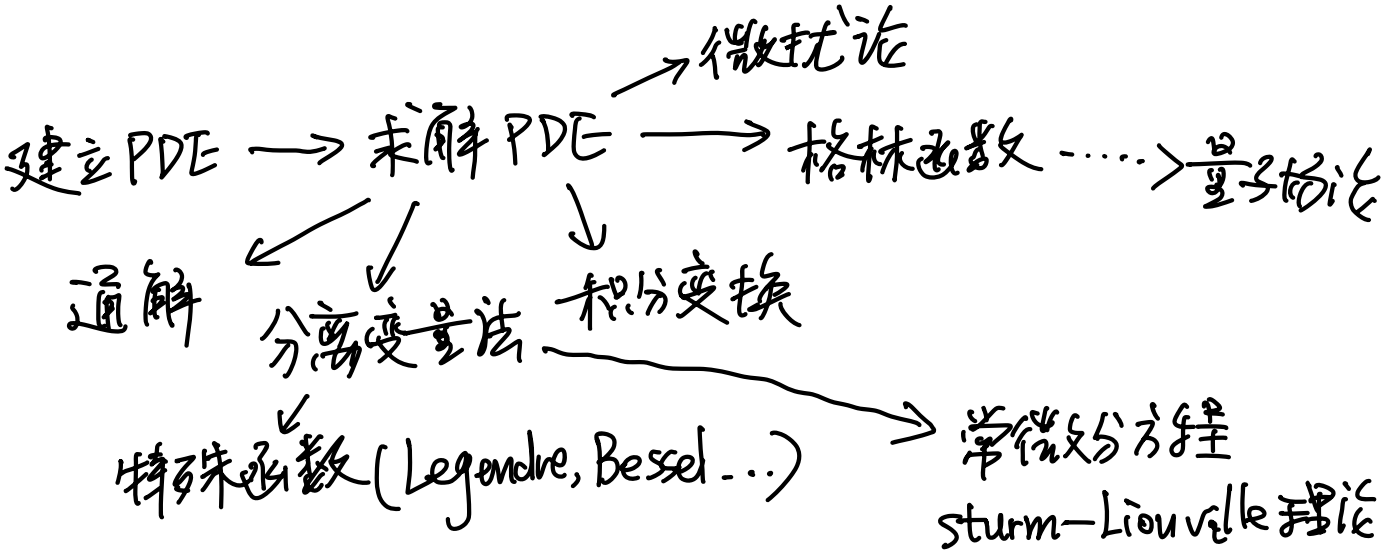
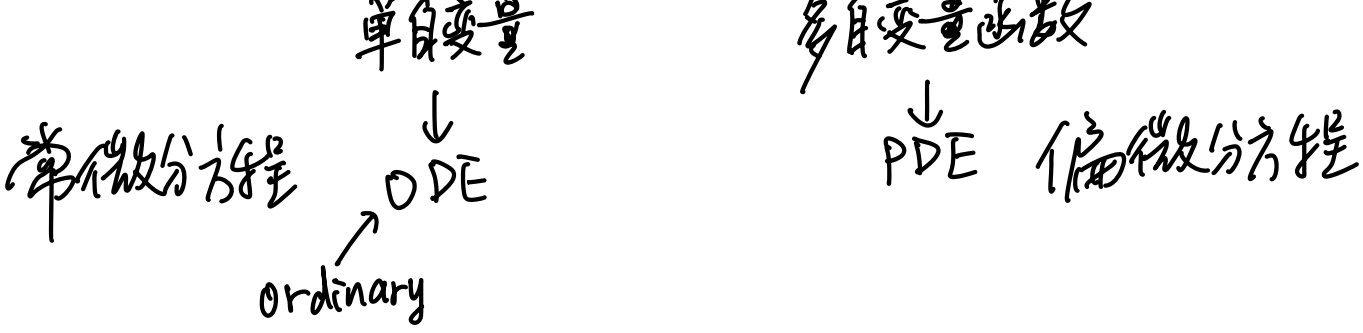
数理方程

第一周
20260227

偏微分方程 partial differential equation PDE





例: $u_t(x,t) + u_x(x,t) = 0$ 单向波方程 (u_t: $\frac{\partial u}{\partial t}$, u_x: $\frac{\partial u}{\partial x}$)

$\nabla^2 u = u_{xx} + u_{yy} + u_{zz} = 0$ Laplace eq.

$\nabla^2 u = \rho(x,y,z)$ Poisson eq.

$u_{tt} - a^2 u_{xx} = 0$

$u_{tt} - a^2 \nabla^2 u = 0$ > wave eq.

$u_t + u u_x = 0$ 激波 shock wave

$\frac{1}{c^2}\Phi_{tt} - \nabla^2 \Phi = \frac{\rho}{\varepsilon_0}$, $\frac{1}{c^2}\vec{A}_{tt} - \nabla^2 \vec{A} = \mu_0 \vec{J}$. Maxwell eq.

$u_t + u u_x + u_{xxx} = 0$ (KdV 孤子)

$u_{tt} + u_{xxxx} = 0$ 梁的振动

$u_t = u_{xx}$ 热传导/扩散方程

$i u_t = -\frac{\hbar^2}{2m}\nabla^2 u + V(x,y,z)u$ schrodinger eq.

$i\gamma^\mu \partial_\mu \psi = m\psi$ Dirac eq.

$u_t = \nabla \cdot (g(|\nabla u|)\nabla u)$ Perona-Malik eq. 图像降噪

AI 扩散模型

帮助理解周围世界

例：① 发令枪之清脆 对比 水波拖尾，奇/偶维空间波动方程解之区别

② 海啸速度→大约飞机速度 浅水波方程

③ 100°C铁和玻璃 烫手程度？ $T_m = \frac{e_1 T_1 + e_2 T_2}{e_1 + e_2}$

effusivity 热渗透率

④ 地下温度季节波动，季节反转

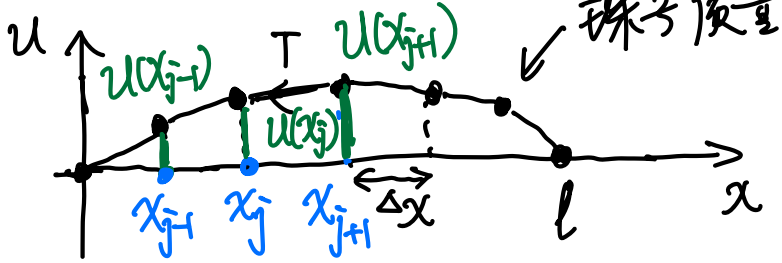
⑤ 秋王马条纹，"图灵斑图"。 反应-扩散方程

建立PDE
- 弦的横振动
- 杆的纵振动
- 热传导/扩散
- 稳态方程

定解条件：
- 初始条件 IC
- 边界条件 BC

弦的横振动：波动方程

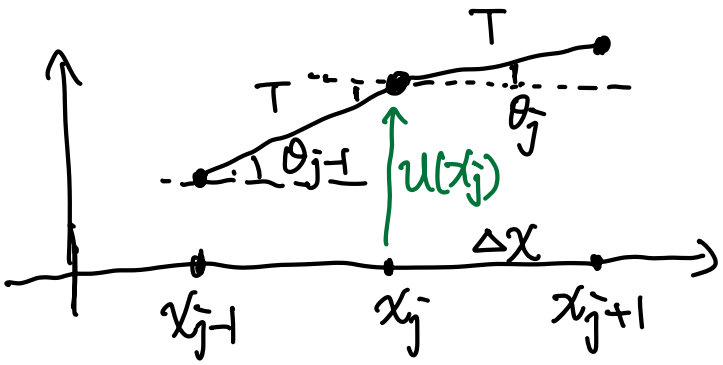
1730s d'Alembert, Euler...



珠子质量 m. 轻弦张力 T

$\Delta x = \frac{l}{N}$ ← 珠子数量

$\Delta x \to 0$ 均匀弦



$T\cos\theta_j - T\cos\theta_{j-1} \approx 0$

≈ 1 ≈ 1

$$mu_{tt}(x_j) = T\sin\theta_j - T\sin\theta_{j-1} \approx T(\tan\theta_j - \tan\theta_{j-1})$$

$$= T\left(\frac{u(x_{j+1})-u(x_j)}{\Delta x} - \frac{u(x_j)-u(x_{j-1})}{\Delta x}\right)$$

忽略$(\theta_j)^2$级

$$= \frac{T}{\Delta x}\left[u(x_{j+1}) + u(x_{j-1}) - 2u(x_j)\right] \to u(x)$$

$x_{j+1} \to x+\Delta x$, $x_{j-1} \to x-\Delta x$, $x_j \to x$

$$u(x+\Delta x) = u(x) + u_x\Delta x + \frac{u_{xx}}{2}\Delta x^2 + \cdots$$

$$u(x-\Delta x) = u(x) - u_x\Delta x + \frac{u_{xx}}{2}\Delta x^2 + \cdots$$

$$= \frac{T}{\Delta x} u_{xx}\Delta x^2$$

$$= T\Delta x u_{xx}$$

$$m u_{tt}(x,t) = T \Delta x\, u_{xx}(x,t)$$

$$\Rightarrow \frac{m}{\Delta x} u_{tt} = T u_{xx} \qquad \Rightarrow \rho u_{tt} = T u_{xx}$$

ρ 线密度

$$\Rightarrow u_{tt} = a^2 u_{xx}, \quad a = \sqrt{\frac{T}{\rho}}$$

一维波动方程　　速度量纲

如果单位长度加横向外力 $F(x,t)$

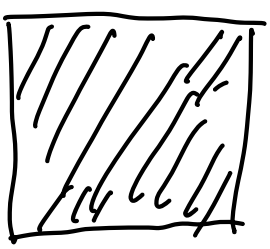


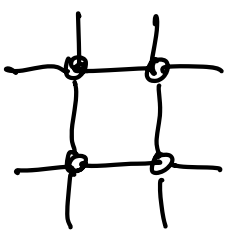
$$\Rightarrow u_{tt} = a^2 u_{xx} + f(x,t)$$

$$f(x,t) = \frac{F(x,t)}{\rho}$$

（自行推导）

推广：二维均匀弹性膜





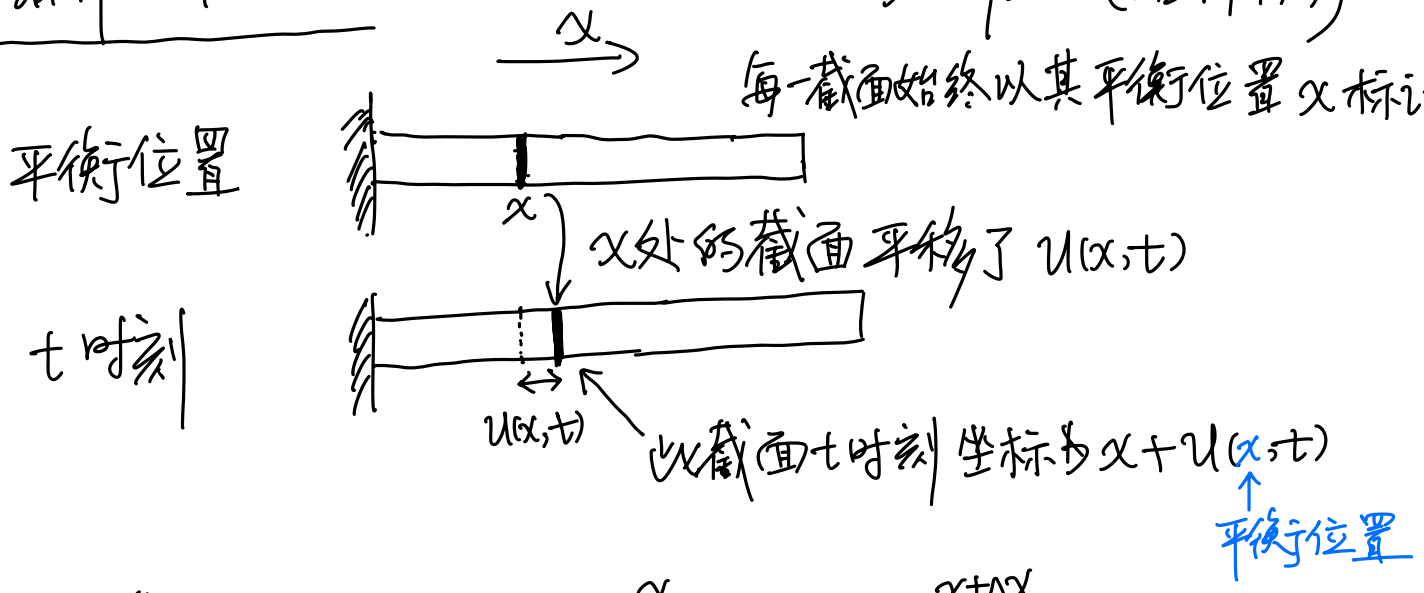
二维波动方程　$u_{tt} = a^2(u_{xx} + u_{yy})$

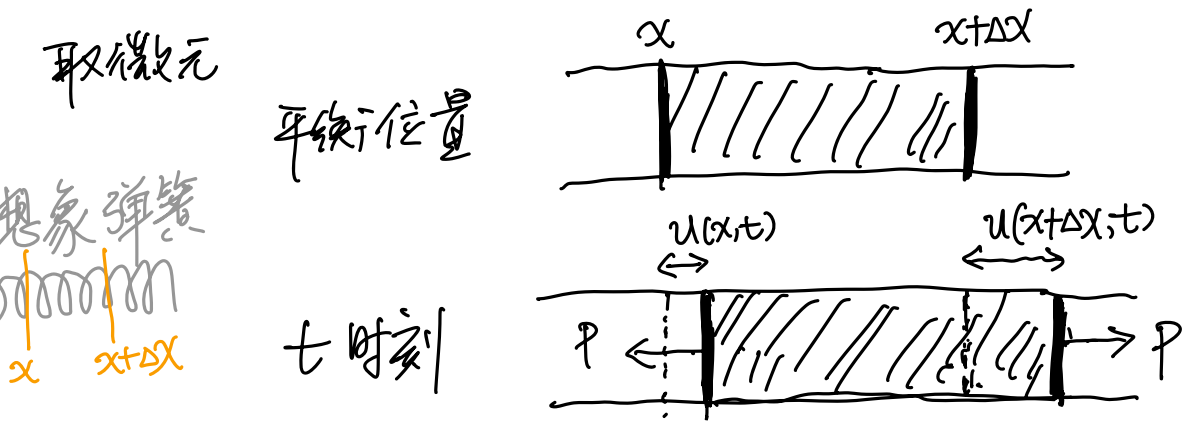


三维波动方程　$u_{tt} = a^2(u_{xx} + u_{yy} + u_{zz})$

声波，电磁波 …

均匀细杆纵振动方程 ⟷ 纵振动(沿杆方向)





单位面积弹性力 $P = E\frac{\partial u}{\partial x}$

E: 材料的杨氏模量

$\frac{\partial u}{\partial x} > 0 \Rightarrow$ 拉伸 $\Rightarrow P > 0$

$\frac{\partial u}{\partial x} = 0 \Rightarrow$ 平衡 $\Rightarrow P = 0$

$\frac{\partial u}{\partial x} < 0 \Rightarrow$ 压缩 $\Rightarrow P < 0$

单位截面积、右侧对左侧的拉力为P，以向右为正. 左侧对右侧的拉力为 $-P$

合力 $\underbrace{P(x+\Delta x)S - P(x)S}_{\text{截面积}} = \underbrace{\rho \frac{S\Delta x}{\text{体积}}}_{\text{体密度}} \frac{\partial^2 u}{\partial t^2}$

$= ES\left(\left.\frac{\partial u}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial u}{\partial x}\right|_{x}\right)$

$$ES\frac{\partial^2 u}{\partial x^2}\Delta x = \rho S\Delta x\frac{\partial^2 u}{\partial t^2}$$

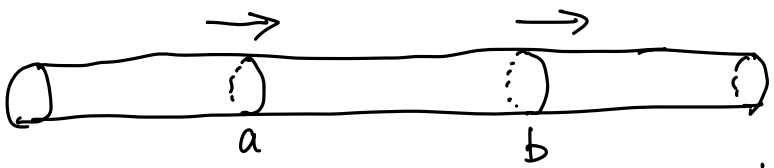
$$\Rightarrow \rho\frac{\partial^2 u}{\partial t^2} = E\frac{\partial^2 u}{\partial x^2} \quad \Rightarrow \quad \frac{\partial^2 u}{\partial t^2} - a^2\frac{\partial^2 u}{\partial x^2} = 0, \quad a = \sqrt{\frac{E}{\rho}}$$

扩散方程　粒子浓度如何随时会变化

一维



浓度 $u(x,t)$ = 单位长度内粒子数



$$\frac{d}{dt}\int_a^b u(x,t)\,dx = q(a,t) - q(b,t) + \int_a^b f(x,t)\,dx$$

↑ 左边流入　↑ 右边流出　↙ 产生率

$\forall a,b$, $[a,b]$不必很短

$q(x,t)$ 扩散流 = 单位时间向右穿过截面的净粒子数

$$\int_a^b \frac{\partial u}{\partial t}dx = -\int_a^b \frac{\partial q}{\partial x}dx + \int_a^b f(x,t)\,dx$$

$$\int_a^b \left(\frac{\partial u}{\partial t} + \frac{\partial q}{\partial x} - f\right)dx = 0 \qquad \forall a,b,$$

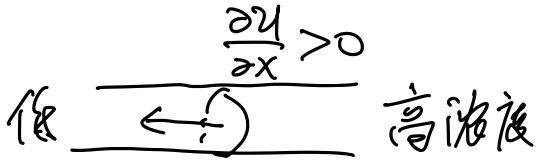
↑ 函数在任意区间上积分 $=0 \Rightarrow$ 函数 $=0$

$$\Rightarrow \frac{\partial u}{\partial t} + \frac{\partial q}{\partial x} - f = 0$$

↑ 从 u 表示 q 即可得 u 的 PDE

Fick's law: $q = -D\frac{\partial u}{\partial x}$

$D>0$ 扩散率



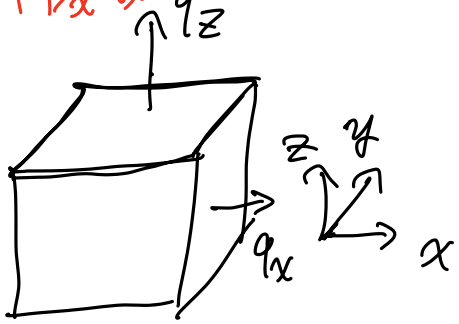
$$\Rightarrow \frac{\partial u}{\partial t} = -\frac{\partial q}{\partial x} + f = \frac{\partial}{\partial x}\left(D\frac{\partial u}{\partial x}\right) + f$$

$D = \text{const} \Rightarrow \frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + f$ 扩散方程

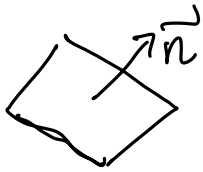
$D = D(x) \Rightarrow \frac{\partial u}{\partial t} = D(x)\frac{\partial^2 u}{\partial x^2} + f$

$\frac{\partial}{\partial x}\left(D(x)\frac{\partial u}{\partial x}\right) = Du_{xx} + D_x u_x$

三维: $u(x,y,z,t)$ 各向同性



$$\vec{q} = -D\nabla u = -D\left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$$



$|\vec{n}| = 1$ 扩散流 $= \vec{q}\cdot\vec{n} = -D\vec{n}\cdot\nabla u$

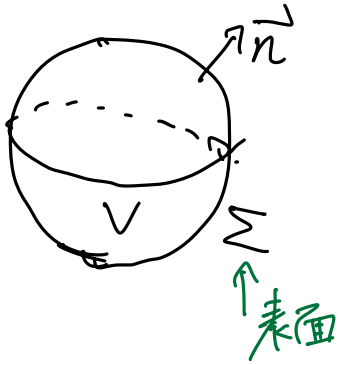
$= -D\left(n_x\frac{\partial u}{\partial x} + n_y\frac{\partial u}{\partial y} + n_z\frac{\partial u}{\partial z}\right)$

$$\frac{d}{dt}\iiint_V u\,dV = -\iint_\Sigma \vec{q}\cdot\vec{n}\,ds + \iiint_V f\,dV$$

dV ↑ $dx\,dy\,dz$

$$= -\iiint \nabla\cdot\vec{q}\,dV \quad \text{divergence theorem}$$

$$= \iiint \nabla\cdot(D\nabla u)\,dV$$



$$\frac{\partial u}{\partial t} = \nabla\cdot(D\nabla u) + f = \frac{\partial}{\partial x}\left(D\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(D\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial u}{\partial z}\right) + f$$

$D = \text{常数} \Rightarrow u_t = D(u_{xx} + u_{yy} + u_{zz}) + f$

$= D\nabla^2 u + f$

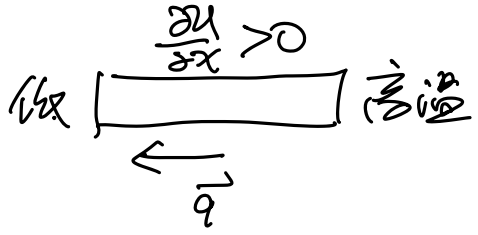
↑ Laplacian 拉普拉斯算符/算子

各向异性：$q_i = -\sum_j D_{ij} \frac{\partial u}{\partial x_j}$ $i, j = x, y, z$.

热传导方程　　热传导 = 能量（分子动能）扩散 ⇒ 扩散方程

傅里叶定律 $\vec{q} = -k \nabla u = -k\left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$

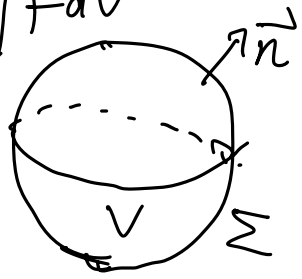
$\vec{q}$：热流密度　k：热导率 thermal conductivity



内能变化 = 净流入热量 + 热产生率

$$\frac{d}{dt}\iiint_V \rho c u \, dV = -\iint_\Sigma \vec{q} \cdot \vec{n} \, dS + \iiint_V F \, dV$$

ρ：密度　c：热容　u：温度



$$-\iiint_V \nabla \cdot \vec{q} \, dV = \iiint_V \nabla \cdot (k \nabla u) \, dV$$

$$\rho c \frac{\partial u}{\partial t} = \nabla \cdot (k \nabla u) + F$$

$$= \frac{\partial}{\partial x}\left(k \frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial z}\left(k \frac{\partial u}{\partial z}\right) + F$$

k = 常数　$\frac{\partial u}{\partial t} = \frac{k}{\rho c} \nabla^2 u + \frac{F}{\rho c} = K \nabla^2 u + f$

$K = \frac{k}{\rho c}$ 热扩散率 thermal diffusivity

⇒ $\boxed{\frac{\partial u}{\partial t} = K \nabla^2 u + f \quad \text{(有热源)}}$

$\boxed{\frac{\partial u}{\partial t} = K \nabla^2 u \quad \text{(无热源)}}$

稳态方程 u不依赖t

1) 扩散方程 $\Rightarrow$ Poisson eq. ↙ 粒子产生/消灭

$$0=\frac{\partial u}{\partial t}=D\nabla^2 u+f(x,y,z)$$

$$\Rightarrow \nabla^2 u=-\frac{f(x,y,z)}{D}$$ Poisson eq.

$$f(x,y,z)=0 \Rightarrow \nabla^2 u=0,$$ Laplace eq.

2) 静电场 $\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon_0}$ $\quad \vec{E}=-\nabla u=\left(-\frac{\partial u}{\partial x},-\frac{\partial u}{\partial y},-\frac{\partial u}{\partial z}\right)$

$$\Rightarrow \nabla^2 u=-\frac{\rho}{\varepsilon_0}$$

3) 波动方程 + 周期外力 $\rightarrow$ Helmholtz eq.

$$\frac{\partial^2 u}{\partial t^2}-a^2\nabla^2 u=f(x,y,z)\cos(\omega t)$$

设解有形式 $u=v(x,y,z)\cos(\omega t)$

$$\Rightarrow [-\omega^2 v(x,y,z)-a^2\nabla^2 v]\cos\omega t=f(x,y,z)\cos\omega t$$

$$\Rightarrow -\omega^2 v(x,y,z)-a^2\nabla^2 v=f(x,y,z)$$

$$\nabla^2 v+k^2 v=-\frac{f}{a^2}$$

$$k=\frac{\omega}{a}$$

$$f=0 \Rightarrow \nabla^2 v+k^2 v=0$$ Helmholtz eq.

驻波,

定态(量子力学)

定解条件：初始条件 + 边界条件
IC BC

初始条件：

波动方程 $\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x,t)$

对 t 二阶。初位移。初速度

$u(x,t)\big|_{t=0} = \phi(x)$, $\frac{\partial u}{\partial t}\big|_{t=0} = \psi(x)$

扩散方程 $\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + f(x,t)$

对 t 一阶。 $u(x,t)\big|_{t=0} = \phi(x)$

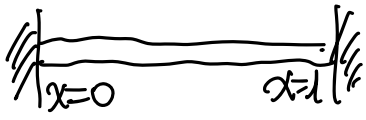
稳态方程无初始条件。

边界条件

三类
- 第一类：指定边界上的函数值 (Dirichlet)
- 第二类：指定边界函数微商 (Neumann)
- 第三类：混合边界条件，指定边界函数值与微商的线性关系 (Robin)

以外：无穷远条件
有界条件
内部边界的连接条件
> 遇见时再谈

例：杆的纵振动

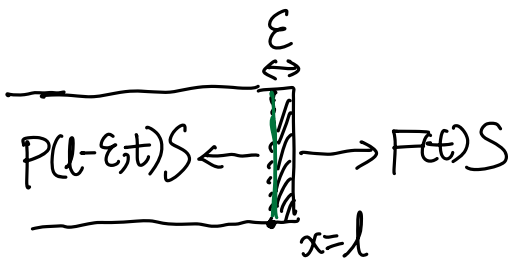


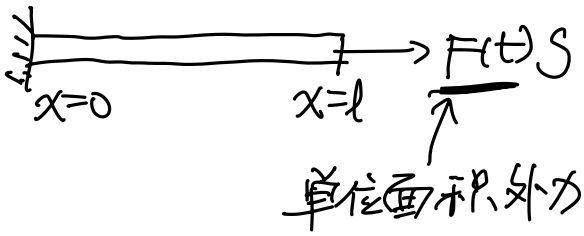
第一类：$u(x,t)\big|_{x=0} = 0$ $u(x,t)\big|_{x=l} = 0$

$u|_{x=0} = A(t)$, $u|_{x=l} = B(t)$ 第一类边界条件

↑ 指定函数

第二类:





$$F(t)S - P(l-\varepsilon,t)S = \rho S\varepsilon \frac{\partial^2 u}{\partial t^2} \xrightarrow{\varepsilon\to 0} 0$$

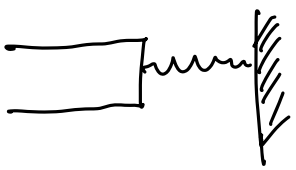
$$\Rightarrow F(t)S = P(l-\varepsilon,t)S = ES\frac{\partial u}{\partial x}\Big|_{x=l-\varepsilon} \approx ES\frac{\partial u}{\partial x}\Big|_{x=l}$$

$$\Rightarrow F(t) = E\frac{\partial u}{\partial x}\Big|_{x=l}$$ ← 右端外力与弹性力平衡

$$\frac{\partial u}{\partial x}\Big|_{x=l} = \frac{F(t)}{E}$$ 第二类

$F(t)=0 \Rightarrow \frac{\partial u}{\partial x}\Big|_{x=l} = 0$ 自由端

第三类:



$$F(t)S = -k[u(x,t) - u_0]_{x=l}$$

↑ $u|_{x=l} = u_0$ 时，弹簧处于平衡位置

$$\Rightarrow \frac{\partial u}{\partial x}\Big|_{x=l} = \frac{F(t)}{E} = -\frac{k}{ES}[u(x,t) - u_0]_{x=l}$$

$$\left(\frac{\partial u}{\partial x} + \frac{k}{ES}u\right)_{x=l} = \frac{k}{ES}u_0$$ 第三类

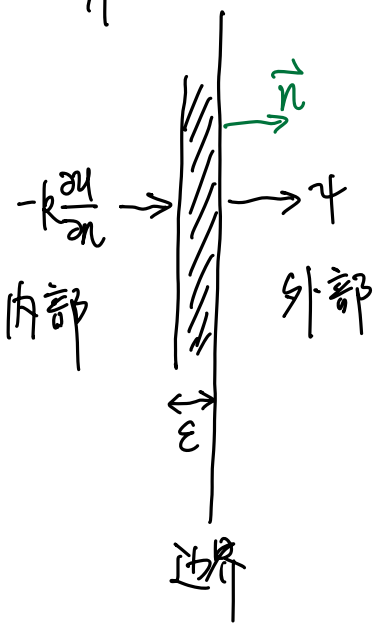
例: 热传导. $\frac{\partial u}{\partial t} = k\nabla^2 u$



第一类：指定边界温度 $u|_\Sigma = \phi(\Sigma, t)$

第二类：指定边界上单位面积流出热量 $\psi(\Sigma, t)$

贴边界取薄层，厚度为 ε.



$\varepsilon \to 0 \Rightarrow$ 热量流入 = 流出

$$-k\frac{\partial u}{\partial n}\Big|_\Sigma = \psi(\Sigma, t)$$

$$\frac{\partial u}{\partial n} = \vec{n}\cdot\nabla u = n_x\frac{\partial u}{\partial x} + n_y\frac{\partial u}{\partial y} + n_z\frac{\partial u}{\partial z}$$

$\frac{\partial u}{\partial n}$：法向微商　$\vec{n}$：向外法线 $|\vec{n}| = 1$

球面：$\frac{\partial u}{\partial n} = \frac{\partial u}{\partial r}$　(r, θ, ϕ)　r：半径

边界绝热 $\psi = 0 \Rightarrow \frac{\partial u}{\partial n}\Big|_\Sigma = 0$

第三类：流出热量依赖于边界温度

$$\psi|_\Sigma = H(u|_\Sigma - u_0)$$

$H > 0$

牛顿冷却定律　u_0：外界温度

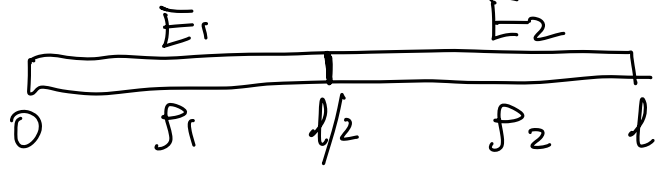
$$\Rightarrow -k\frac{\partial u}{\partial n}\Big|_\Sigma = \psi|_\Sigma = H(u|_\Sigma - u_0)$$

$$\Rightarrow \left(\frac{\partial u}{\partial n} + \frac{H}{k}u\right)_\Sigma = \frac{H}{k}u_0, \quad \left(\frac{\partial u}{\partial n} + hu\right)_\Sigma = hu_0$$

$h = \frac{H}{k}$

连接条件：内部性质出现跃变，跃变处附加条件

例：



① $u|_{x=\frac{l}{2}-0} = u|_{x=\frac{l}{2}+0}$ 两侧位移相等（否则杆就断了）

② $E_1 S \frac{\partial u}{\partial x}\Big|_{\frac{l}{2}-0} = E_2 S \frac{\partial u}{\partial x}\Big|_{\frac{l}{2}+0}$ 弹性力相等.

$\Rightarrow E_1 \frac{\partial u}{\partial x}\Big|_{\frac{l}{2}-0} = E_2 \frac{\partial u}{\partial x}\Big|_{\frac{l}{2}+0}$

PDE + 定解条件（初始 + 边界）$\Longrightarrow$ 定解问题

适定性：〈适定 well posed
〈不适定 ill posed

适定问题：其解具有

1) 存在性：条件不能过多

例如热传导方程给定 $u|_{t=0}$ 和 $\frac{\partial u}{\partial t}|_{t=0} \Rightarrow$ 无解

2) 唯一性 条件不能过少，否则解不唯一

3) 稳定性 定解条件作微小改变时，解也只有微小改变. 例如，波动方程的初位移和初速度作微小改变，$u(x,t)$ 也只作微小改变

物理量的测量值与真实值难免有微小误差，若解无稳定性，其意义要小心对待.

定解条件

不适定问题并非全无意义，有些不适定问题也能得出有价值的解，后面将通过例子说明.